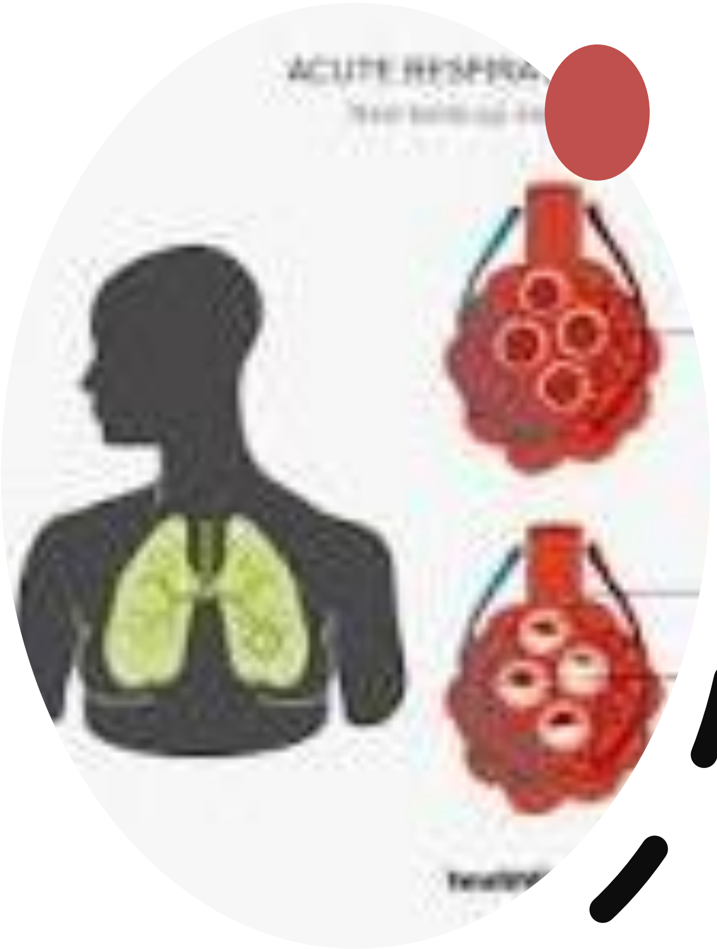
Respiratory failure



Prepared by :

Dr / Ghada Elsayed

outlines

- Introduction
- Pathophysiology
- Causes of Acute Respiratory Failure
- Classification
- Assessment
- Diagnostic studies
- Complication
- Managemnt of RF

Introduction

Acute respiratory failure

is a sudden and life-threatening deterioration in pulmonary gas exchange, resulting in carbon dioxide retention and inadequate oxygenation.

- Acute respiratory failure remains a major cause of morbidity and mortality in the intensive care setting

Causes of Acute Respiratory Failure

- **Intrinsic Lung/Airway Diseases:**
- Large Airway Obstruction: Foreign bodies.
- Bronchial Diseases: (Chronic bronchitis, Asthma).
- Parenchymal Diseases: (Pulmonary emphysema, severe pneumonia).
- Cardiovascular Disease: Cardio pulmonary edema.

Extrapulmonary Disorders

- Diseases of the Pleura and the Chest Wall:

Pneumothorax,, Traumatic injury to the chest wall, flail chest.

-Disorders of the Peripheral Nerves and Spinal Cord:

Poliomyelitis, Guillain-Barré syndrome, Spinal cord trauma

-Disorders of the Central Nervous System:

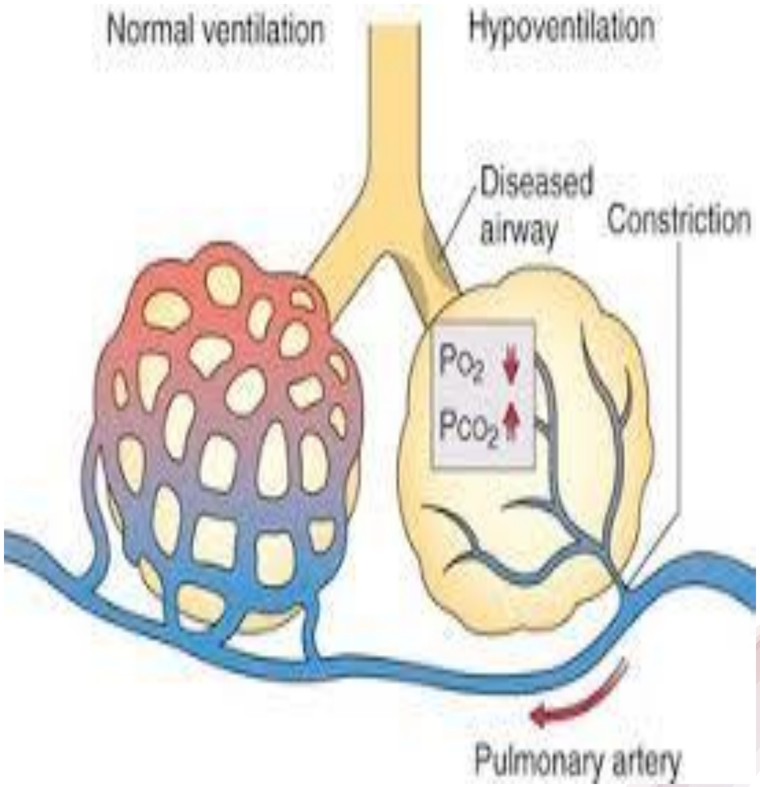
Sedative and narcotic drug overdose, Head trauma, Cerebrovascular accident, Sleep apnea syndrome.

Pathophysiology of Respiratory Failure

- Hypoxemia is the result of impaired gas exchange and is the **hallmark** of ARF.
- The main *causes of hypoxemia are:*
- **Alveolar hypoventilation**
- **Ventilation/perfusion (V/Q) mismatching**
- **Intrapulmonary shunting**

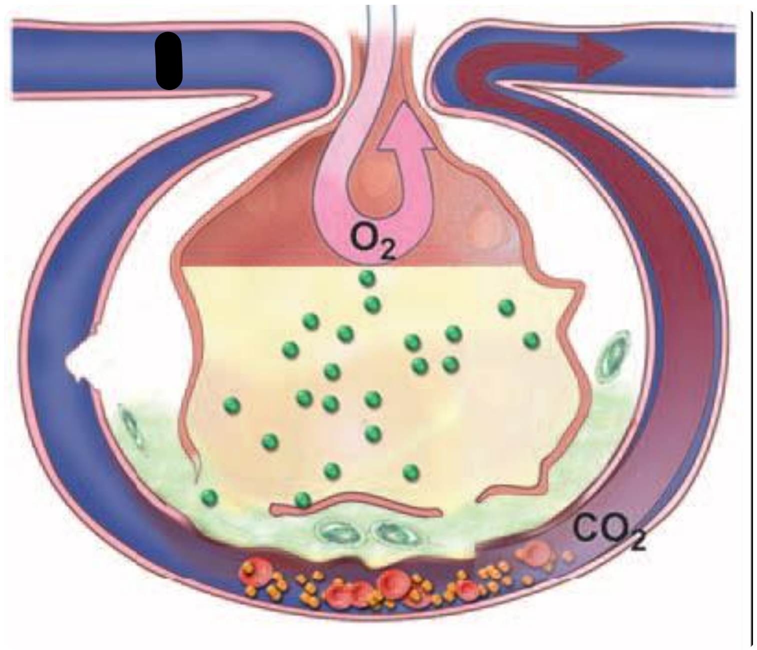
1. Alveolar hypoventilation

It occurs when the amount of oxygen being brought into the alveoli is insufficient to meet the metabolic need of the body. This can be the result of increasing metabolic oxygen needs or decreasing ventilation. Hypoxemia caused by alveolar hypoventilation is associated with hypercapnia and commonly results from extra-pulmonary disorders.



V/Q mismatching

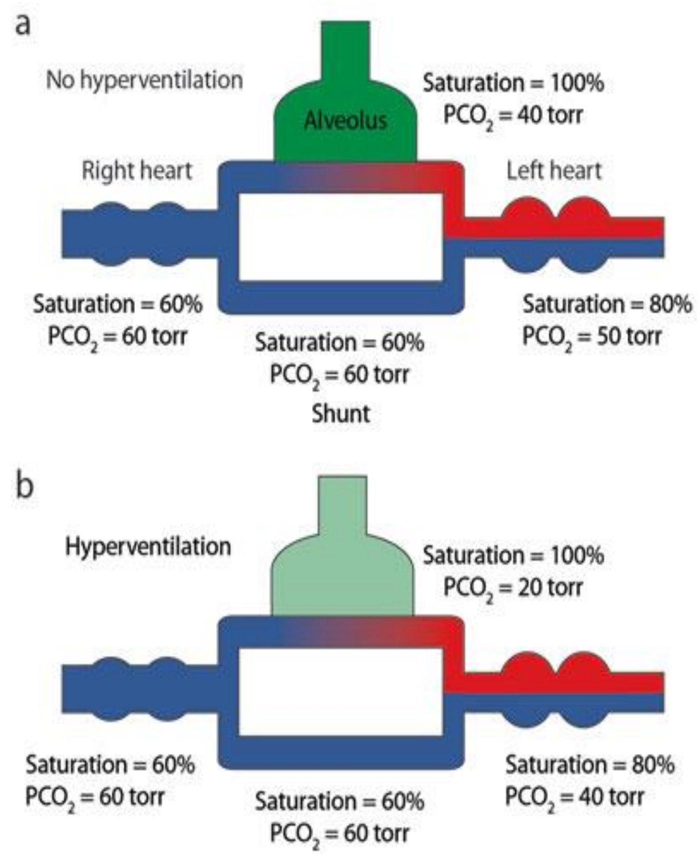
- Blood passes through alveoli that
- are under ventilated for the given amount of perfusion.
- The blood leaving these areas has a lower than normal amount of oxygen . (Alveoli are partially collapsed or partially filled with fluid)



Intrapulmonary shunting

➓ Intrapulmonary shunting

occurs when blood passes through a portion of a lung that is not ventilated, which may be the result of alveolar collapse(e. g, atelectasis), alveolar consolidation (e.g., pneumonia), or excessive mucus accumulation (e.g., chronic bronchitis).



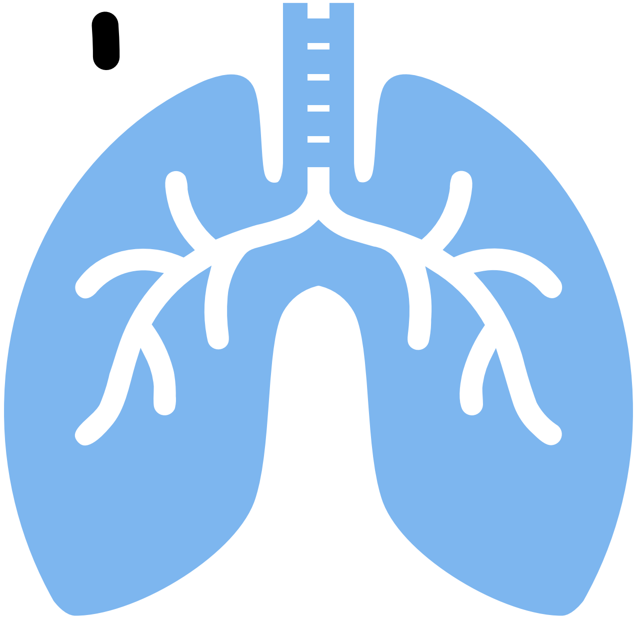
Classification

1-Acute Hypoxemic Respiratory Failure: type I

inability to achieve adequate oxygenation, PaO_2 of 50 mm Hg or less and $PaCO_2$ 40 mm Hg or less.

2-Acute hypercapnic respiratory failure: Type II

inadequate alveolar ventilation and is characterized by marked elevation of carbon dioxide with relative preservation of oxygenation



Classification of Respiratory Failure

Type I Respiratory Failure: Hypoxemic normocapnic failure or oxygenation failure, the body fails to deliver an adequate supply of oxygen to the arterial blood. The patient presents with a low PaO2 less than 60 mmHg and a normal PaCO2.

Type II Respiratory Failure: Hypoxemic hypercapnic failure or ventilatory failure, the body ineffectively eliminates carbon dioxide (CO_2) from the arterial blood. The patient presents with a low PaO_2 less than 60 mmHg and $PaCO_2$ is greater than 50 mmHg and pH is less than or equal 7.30.

Assessment

- **HISTORY**
- A complete medical and social history should be obtained from the patient or a family member to determine the patient's baseline respiratory status on admission.
- assessment of adult patients with actual or potential pulmonary dysfunction.

History and Symptoms Profile

PULMONARY SYMPTOMS*

-DYSPNEA

- COUGH,

-SPUTUM,

-WHEEZE,

- CHEST PAIN .

EXTRAPULMONARY SYMPTOMS*

-NIGHT SWEATS,

-HEADACHE ON AWAKENING,

-FLUID RETENTION

-FATIGUE,

-PAROXYSMAL NOCTURNAL APNEA

-SLEEP DISTURBANCES,

-SINUS PROBLEMS

Previous History

PULMONARY PROBLEMS

- **TREATMENTS**
- **NUMBER OF HOSPITALIZATIONS**
- **MEDICAL DIAGNOSIS**
- **IMMUNIZATION**

PHYSICAL FINDINGS

-TYPE ONE RF (Hypoxemia)

-Respiratory (dyspnea, tachypnea, prolonged expiration, use accessory respiratory, cyanosis).

-Cerebral (restlessness, confusion, coma)

-Cardiac (tachycardia, arrhythmia, hypertension, hypotension).

-Fatigue, unable to speak.

Type II RF . **Hypercapnia**

- **Respiratory (dyspnea)**
- **Cerebral (morning headache, disorientation, coma)**
- **Cardiac (dysrrhythmia)**
- **Neuromuscular (muscle weakness, tremors)**
- **Pursed lip breathing.**

Diagnostic studies

- **ABG:**

to determine the exact level of **Pa02, PaCO2**, and blood **pH** in cases of suspected (ARF).

Only determination of the blood gases and pH can confirm the diagnosis.

- **Chest radiography.**
- **Sputum culture.**
- **Pulmonary function tests.**
- **Angiography.**

- **V/Q scanning.**
- **CT**
- **CBC**
- **Serum electrolytes.**
- **Urine analysis.**
- **Bronchogram.**
- **Bronchoscopy.**
- **ECG**
- **Echcardiography.**

Management of Acute Respiratory Failure

Aimed to:

- **Correct underlining cause**
- **Promoting adequate gas exchange**
- **Correcting acidosis**
- **Initiating nutrition support**
- **Preventing complications**

1. Oxygenation

Improve oxygenation through :

- Supplemental oxygen administration, either with a low flow system or a high flow system
- Use of positive airway pressure to correct hypoxemia within the arterial hemoglobin oxygen saturation greater than 90% without cause hypercapnia or oxygen toxicity.
- Positive pressure is necessary to open collapsed alveoli and facility their participation in gas exchange. Positive pressure is delivered by invasive or non invasive (by the mask) mechanical ventilation.

2. Ventilation

- Intervention to improve ventilation includes the use of invasive and noninvasive mechanical ventilation depending on causes and severity of disease. Patients with a PH less than 7.25 need for invasive mechanical ventilation and started on volume ventilation in the assist/ control mode.

3- Nutrition Support

The goal of nutrition support are:

- Meet the overall nutrition needs of the patient while avoiding overfeeding.
- Avoid nutrition delivery related complication and
- Improve patient outcomes.

- **4-Restoration of fluid and electrolyte balance**

1-Prevent excessive intravenous fluid or conversely, poor fluid intake.

2-Monitor fluid intake and output closely.

3-Perform daily body weight measurement.

4-Prevent and treat promptly hypokalemia and hypophosphatemia.

- **5-Optimization of cardiac function**

1-Maintain adequate cardiac output.

2-use of pulmonary artery catheter for accurate hemodynamic monitoring.

treatment of underlying correctable conditions and precipitating causes

- **Prevent or treat respiratory tract infections**
- **Prevent potential airway obstruction**
- **Identify and treat congestive heart failure appropriately.**
- **Recognize bronchospasm with bronchodilators and corticosteroids.**
- **Assess tolerance to sedative, hypnotic, and narcotic drugs**

- **Remove air or fluid in the pleural cavity.**
- **Prevent and treat abdominal distension by insertion of a nasogastric tube.**
- **assess limitation of the thoracic wall movement,**

ineffective cough, and lack of deep breathing (trauma surgical patients)

- **Assess diaphragmatic fatigue; if present**

Periodic assessment of the response to therapy

- Perform frequent arterial blood gas measurements.
- Monitor arterial oxygen saturation by pulse oximetry.
- Prevention and early detection of potential complications

Complications

- **Ischemic anoxic encephalopathy**: result from hypoxemia, hypercapnia, and acidosis.
- **Dysrhythmias** are precipitated by hypoxemia, acidosis,electrolytes imbalances.
- ***Venous thromboempolism***: precipitated by venous stasis resulting from immobility and can be prevented through the use of intermittent pneumatic compression device and low dose of unfractionated heparin or low molecular weight heparin.

Cont, complication

- ***GIT bleeding*** can be prevented through the use of histamine receptor antagonists, Cytoprotective agent or proton pump inhibitors.
- ***Others complication*** associated with an artificial airways, mechanical ventilation, enteral or parenteral nutrition and peripheral arterial cannulation.

Pharmacological intervention

Medications to facilitate dilation of the airways.

- Bronchodilators, such as beta2- agonists and anticholinergic agents, aid in smooth muscle relaxation and are of particular benefit to patients with airflow limitations.
- Steroids to decrease airway inflammation and enhance the effects of the beta2-agonists.
- Sedation is necessary in many patients to assist with maintaining adequate ventilation. It can be used to comfort the patient and decrease the work of breathing,
- Analgesics should be administered for pain control.

Any question

Thank you